

COURSE REVIEW

Cardiovascular Physiology

Cardiac function, circulation, ECG and blood physiology

Updated 8 October 2026

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SCOPE

Physiology-only selection from the existing Cardiovascular System review. The 27 original physiology and blood sections are preserved with their figures, self-checks and source references. Anatomy, histology, embryology, mixed appendices and transcript attachments are excluded.

How to read this book

	Key idea	The idea to hold on to. Consolidates the section.
	Exam trap	A common wrong answer or transcription error, and the correction.
1	Check yourself	Short retrieval questions with the answer directly below. Cover the answer, then test yourself.

Lecture-grounded material and book extensions are distinguished in each section's source line. This is a detailed review of the available sources, not an official exam blueprint.

Original labelled figures remain important. Numerical laboratory ranges and procedures depend on the course method and local SOP.

This physiology-only extract preserves the selected authored sections and their source references. Shared integrated appendices and lecture transcript attachments are not included. Personal study only: original book and lecture illustrations retain their owners' rights.

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SECTION 01 · LECTURE + BOOK

Cardiac action potentials and excitation-contraction coupling

CELL/PHASE	DOMINANT MECHANISM	EXAMINATION DISTINCTION
Ventricular phase 0	Fast Na ⁺ channel opening and inward Na ⁺ current	Upstroke/conduction differ from nodal calcium-dependent upstroke
Phase 1	Transient outward K ⁺ current after Na ⁺ inactivation	Brief early repolarization
Phase 2 plateau	Inward L-type Ca ²⁺ balanced against outward K ⁺ currents	Sustains contraction and long refractoriness
Phase 3	Ca ²⁺ channels inactivate and outward K ⁺ dominates	Repolarization is K ⁺ efflux, not K ⁺ influx
Phase 4 ventricular	Stable negative potential with high resting K ⁺ conductance	Ordinary working myocytes do not show the normal SA pacemaker slope
SA/AV phase 4	Funny current, declining K ⁺ efflux and Ca ²⁺ contributions	Spontaneous diastolic depolarization
SA/AV phase 0	L-type Ca ²⁺ influx	Fast Na ⁺ channels are not the principal upstroke mechanism

Depolarization travels down T tubules, opens L-type calcium channels and triggers ryanodine receptor-mediated calcium release from sarcoplasmic reticulum. Calcium binds troponin C, permitting actin-myosin cycling. Relaxation requires SERCA reuptake, sodium-calcium exchange and extrusion. ATP is required both for cross-bridge cycling and restoration of calcium gradients. The long action potential/refractory period prevents fused tetanic contraction and preserves filling intervals.

Beta-1 stimulation increases cAMP-dependent calcium handling and pacemaker activity, raising contractility and rate while accelerating relaxation. Parasympathetic M2 signaling reduces cAMP and increases potassium conductance, slowing nodal firing/conduction. Do not repeat ASR phrases that swap sympathetic with parasympathetic or call outward K⁺ current an influx.

CHECK YOURSELF

1 Why does increased extracellular calcium availability affect cardiac contraction more directly than ordinary skeletal contraction?

Cardiac excitation-contraction coupling depends on calcium entry through L-type channels to trigger SR release. Skeletal muscle uses mechanically coupled voltage sensors and SR release much more directly. Cardiac force therefore depends strongly on the extracellular trigger calcium.

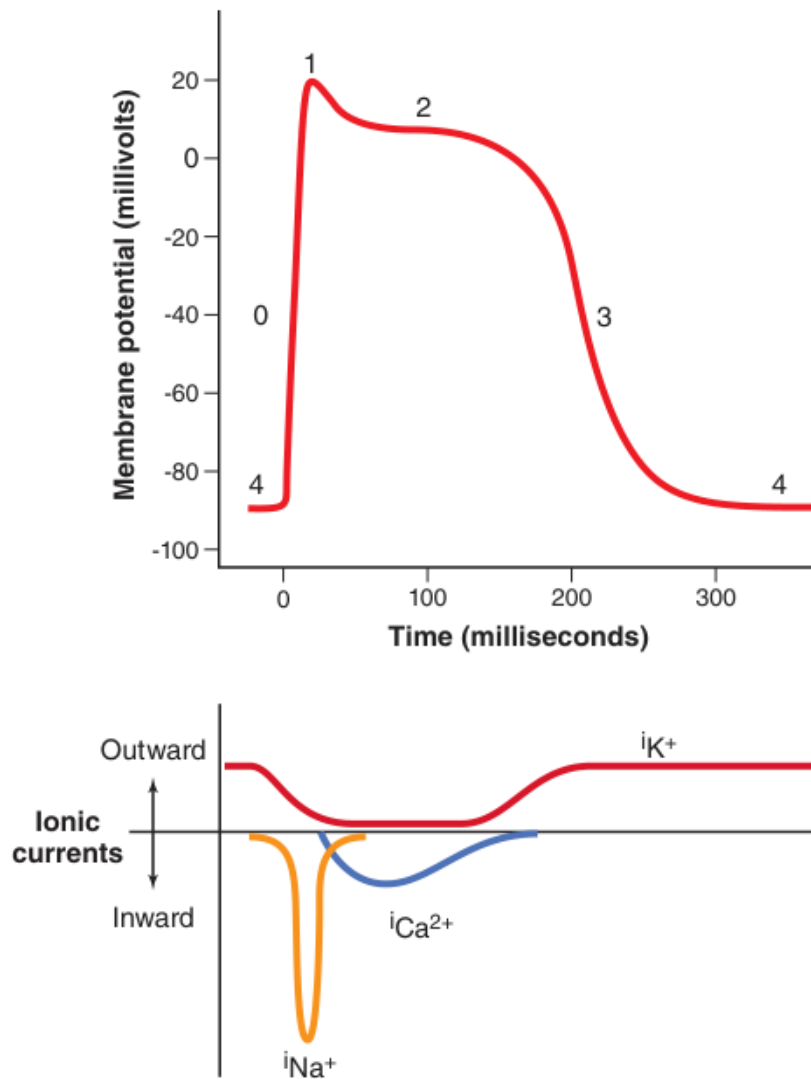


Figure 9.5 Phases of action potential of cardiac ventricular muscle cell and associated ionic currents for sodium (i_{Na^+}), calcium ($i_{Ca^{2+}}$), and potassium (i_{K^+}).

Figure 1 Relate phases 0-4 to fast sodium entry, sustained calcium current and outward potassium current.

Identify Fast Na^+ current; L-type Ca^{2+} current; K^+ current; Plateau

Source: Guyton 2026, Fig. 9.5, PDF p. 123

SOURCES Cardiac action potential with narration.ppt - Slides 1-17 · Guyton and Hall Textbook of Medical Physiology (2026).pdf - PDF pp. 121-125, 135-141

SECTION 02 · LECTURE + BOOK

Pacemakers, AV delay and conduction failure

The SA node normally dominates because its spontaneous depolarization reaches threshold faster than other latent pacemakers. Impulses spread through atria, slow in AV nodal tissue, then traverse His bundle, bundle branches and Purkinje network to activate ventricular myocardium rapidly. AV delay permits atrial contraction to precede ventricular contraction. Purkinje conduction is rapid; AV nodal conduction is relatively slow. The fibrous AV separation forces the normal impulse through this specialized connection.

An ectopic focus or re-entry circuit can compete with the sinus node. Re-entry requires an excitable pathway, unidirectional block and a conduction delay sufficient for tissue ahead to recover excitability. A premature ventricular beat can produce a wide QRS because activation spreads abnormally through ventricular myocardium rather than rapidly through the normal network. AV nodal blockade can expose a slower escape rhythm from subsidiary pacemakers.

CHANGE	LIKELY CONSEQUENCE	REASON
Increased vagal tone	Longer AV delay, slower sinus rate	M2 effects on nodal currents
Sympathetic activation	Faster SA firing and AV conduction	Increased pacemaker slope and calcium current
Bundle branch block	Widened ventricular depolarization	One ventricular region activates later by an alternate route
Accessory AV pathway	Potential pre-excitation/re-entry	Conduction bypasses normal AV nodal timing

EXAM TRAP**Electrical versus mechanical**

A recorded depolarization does not prove an effective pulse. Electrical activation must still generate coordinated calcium-dependent contraction, fill the chamber and eject against the outflow pressure.

SOURCES Rythmical Excitation.ppt - Slides 1-5 · Guyton and Hall Textbook of Medical Physiology (2026).pdf - PDF pp. 135-141

SECTION 03 · LECTURE + BOOK

Cardiac cycle, pressure crossings and heart sounds

Valve motion follows pressure gradients. At the start of ventricular systole, ventricular pressure exceeds atrial pressure and AV valves close. During isovolumetric contraction both inlet and outlet valves of that ventricle are closed, pressure rises and volume remains at EDV. When ventricular pressure exceeds arterial pressure, the semilunar valve opens and ejection begins. At the end of ejection, arterial pressure exceeds ventricular pressure and the semilunar valve closes. Isovolumetric relaxation reduces pressure at ESV until atrial pressure opens the AV valve.

EVENT	ECG/PRESSURE ASSOCIATION	SOUND OR WAVE
Atrial systole	After P wave; augments ventricular filling	Atrial a wave
Ventricular contraction	After QRS; AV valve closure	S1; atrial c wave partly from AV valve bulging

EVENT	ECG/PRESSURE ASSOCIATION	SOUND OR WAVE
Ejection	Ventricle above outflow pressure	Arterial systolic pressure rise
Semilunar closure	Near end of T wave/mechanical systole	S2; aortic incisura/dicrotic notch
Venous filling of closed atrium	During ventricular systole	Atrial v wave
Rapid ventricular filling	After AV valve opens	S3 can accompany rapid filling
Atrial contraction into stiff ventricle	Late diastole	S4 requires effective atrial contraction

P represents atrial depolarization, not atrial contraction itself. QRS precedes ventricular pressure rise, and T reflects ventricular repolarization. At faster rates diastole shortens disproportionately, reducing filling time and coronary perfusion time. Right and left heart valves do not all change state at exactly the same instant. Inspiration normally delays pulmonary closure relative to aortic closure, producing physiologic S2 splitting.

CHECK YOURSELF

1 What happens to ventricular volume between mitral closure and aortic opening?

It remains approximately constant at EDV while pressure rises. The interval is isovolumetric contraction. If an inlet or outlet valve is incompetent, the ideal truly isovolumetric assumption no longer applies.

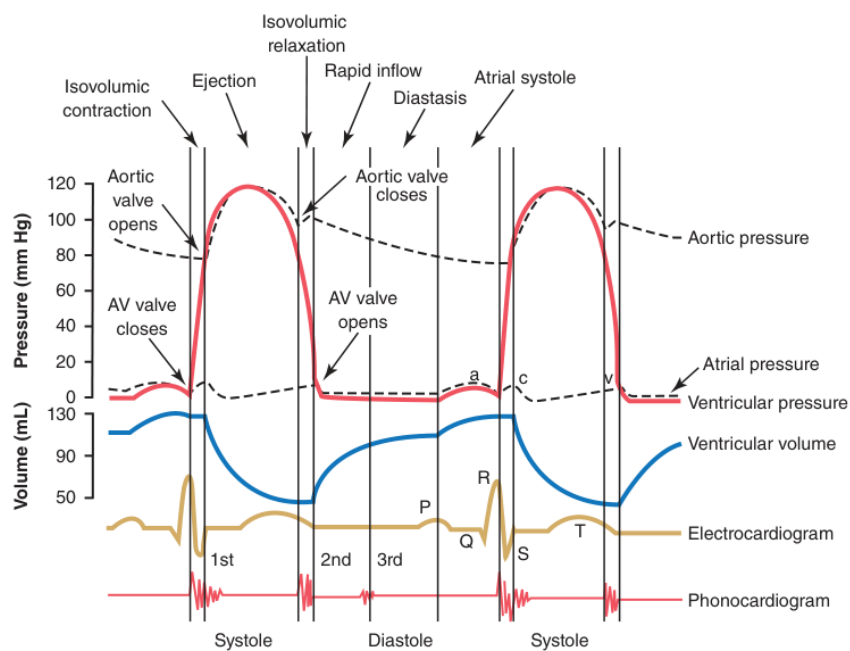


Figure 9.8 Events of the cardiac cycle for left ventricular function, showing changes in left atrial pressure, left ventricular pressure, aortic pressure, ventricular volume, the electrocardiogram, and the phonocardiogram. AV, Atrioventricular.

Figure 2 Read each pressure crossover before identifying valve opening, closure, sound and volume change.

Identify Aortic pressure; Atrial pressure; Ventricular pressure; Ventricular volume; S1; S2

Source: Guyton 2026, Fig. 9.8, PDF p. 126

SOURCES Cardiac Physiology with narration 1.ppt - Slides 1-10 · Guyton and Hall Textbook of Medical Physiology (2026).pdf - PDF pp. 126-130

SECTION 04 · LECTURE + BOOK

Pressure-volume loops, preload, afterload and contractility

Stroke volume is EDV minus ESV, and ejection fraction is stroke volume divided by EDV. A normal teaching example of EDV 120 mL and ESV 50 mL gives SV 70 mL and EF about 58%. Cardiac output is HR times SV, so 70 beats/min times 70 mL is 4.9 L/min. These numbers are illustrative, not universal individual reference intervals. A PV loop is traced counterclockwise: filling, isovolumetric contraction, ejection and isovolumetric relaxation.

CHANGE CONSIDERED IN ISOLATION	PV-LOOP TENDENCY	MECHANISM
Increased preload	Larger EDV and wider loop	More initial fiber stretch increases force through Frank-Starling behavior
Increased afterload	Higher systolic pressure, greater ESV and smaller SV initially	More pressure required before and during ejection
Increased contractility	Lower ESV, steeper end-systolic relation and greater SV	More force at a given preload
Reduced diastolic compliance	Higher filling pressure for a given EDV	Stiffer chamber shifts diastolic pressure-volume relation upward

Preload describes initial myocardial stretch and is approximated by end-diastolic volume/pressure only in context. Afterload is wall stress opposing shortening, not simply blood pressure in every situation. Contractility is intrinsic force generation at a given loading condition, not increased force merely because preload rose. The loop area represents external stroke work. Pressure development and wall tension also consume oxygen even when ejected volume is limited.

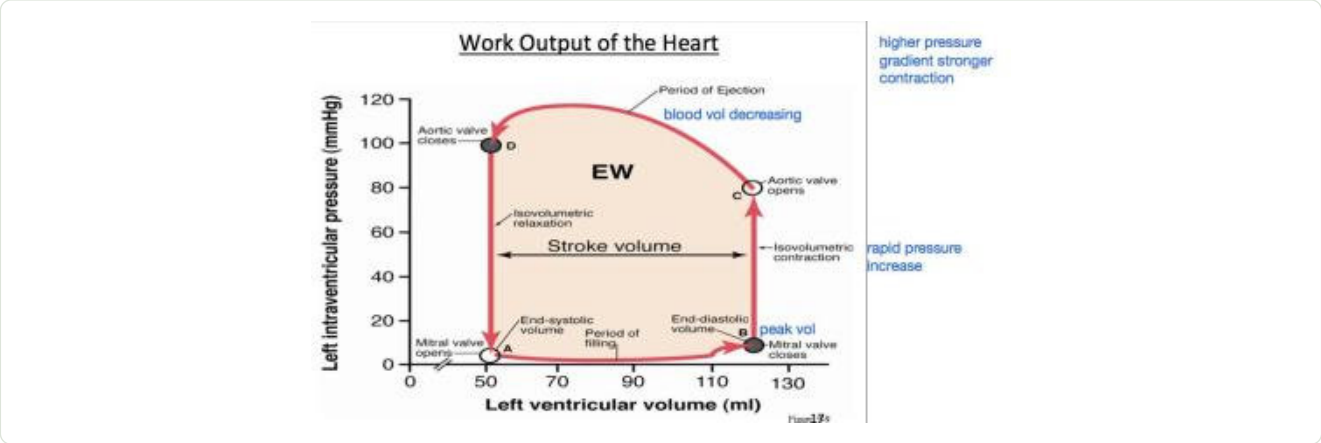


Figure 3 Work through the valve events at each corner before interpreting changes in loop width or height.

Identify Mitral closure; Aortic opening; Aortic closure; Mitral opening; Stroke work

Source: Cardiac physiology with naration 2.pptx, slide 4

CHECK YOURSELF

- 1 **A ventricle has a normal EF but requires an abnormally high filling pressure to reach its EDV. Which property is impaired?**

Diastolic compliance is reduced. EF alone does not measure ease of filling and cannot exclude dysfunction in a stiff ventricle.

SOURCES Cardiac physiology with narration 2.pptx - Slides 3-7 · Guyton and Hall Textbook of Medical Physiology (2026).pdf - PDF pp. 129-134, 250-260

SECTION 05 · LECTURE + BOOK

Pressure, flow, resistance and vascular networks

Flow Q equals pressure difference divided by resistance: $Q = (P_{in} - P_{out})/R$. A high absolute pressure does not imply high flow if downstream pressure rises equally. For systemic circulation, the driving difference is approximately MAP minus right atrial pressure. Pulmonary resistance uses mean pulmonary arterial minus left atrial/wedge pressure, divided by pulmonary flow. Always keep units consistent: pressure in mmHg and flow in L/min yield mmHg·min/L, not the older mmHg·s/mL resistance unit.

For an ideal rigid cylindrical tube with steady laminar Newtonian flow, resistance is $8\eta L/(\pi r^4)$. Doubling radius reduces resistance to one-sixteenth if other factors remain unchanged. Halving radius raises resistance sixteenfold. Real blood is non-Newtonian and vessels are compliant/branching/pulsatile, so this law is a model rather than an exact description of every vessel. Hematocrit strongly affects viscosity. Plasma proteins also contribute. Values depend on temperature and shear rate.

ARRANGEMENT	RESISTANCE RELATIONSHIP	FLOW RELATIONSHIP
Series	$R_{total} = R_1 + R_2 + \dots$	Same steady flow passes sequential components; pressure drops add
Parallel	$1/R_{total} = 1/R_1 + 1/R_2 + \dots$	Branches share an inlet-outlet pressure difference; flows add
Systemic versus pulmonary	Two circuits in series	Average right and left output are equal at steady state
Systemic organs	Largely parallel branches	Each organ regulates its own resistance and share of output

CHECK YOURSELF

- 1 **Three equal vascular beds of resistance 6 units are perfused in parallel. What is total resistance, and what happens if one closes?**

Initially $1/R = 3/6$, giving $R = 2$ units. With two open beds $R = 3$ units. At an unchanged pressure difference, total flow falls to two-thirds of the initial value. Flow through each remaining bed need not fall if its own resistance and driving pressure remain unchanged.

In a thin-walled cylindrical vessel, circumferential tension per unit length is approximately transmural pressure times radius, and wall stress is approximately pressure times radius divided by wall thickness. A thin spherical chamber has a factor of two in the corresponding denominator. These geometric idealizations explain why dilation raises required tension and why thicker walls can reduce stress; tension, stress and pressure are not interchangeable quantities. Compliance is absolute volume change per pressure change, while distensibility divides that volume change by starting volume. The arterial-pressure section applies the distinction to pulse pressure.

SOURCES Cairculatory System 1.pptx - Slides 16-28 · Cairculatory System 2.pptx - Slides 5-10 · Guyton and Hall Textbook of Medical Physiology (2026).pdf - PDF pp. 177-186

SECTION 06 · LECTURE + BOOK

Velocity, Bernoulli, Reynolds and flow measurement

Average velocity equals flow divided by total cross-sectional area, $v = Q/A$. Capillary velocity is low because the many parallel capillaries have a very large aggregate area. A small single vessel does not necessarily have slower velocity if the same flow is forced through it. Across a narrowed segment at fixed flow, velocity rises. By contrast, when pressure gradient is fixed and a tube narrows, resistance rises and flow falls. The held-constant condition determines the correct answer.

Reynolds number is proportional to density $\times$ velocity $\times$ diameter / viscosity. Higher Reynolds numbers favor turbulence. Rapid flow, large diameter and low viscosity can increase the tendency. A stenosis promotes disturbances through high local velocity and geometry despite its reduced diameter. The lecture inequalities on laminar/turbulent flow are reversed: turbulence is associated with higher, not lower, Reynolds number. Thresholds in pulsatile branching vessels are not one universal hard boundary.

Bernoulli relates static pressure, kinetic energy per volume and gravitational potential for ideal flow along a streamline. Higher velocity can accompany lower local static pressure, but real vascular flow also loses energy through viscous resistance. Hydrostatic pressure is ρgh relative to a reference level. Mean systemic filling pressure is a volume/tone-dependent pressure when flow ceases, not simply another name for ρgh . Pulse-wave speed is propagation of a pressure disturbance and differs from the actual velocity of blood.

- An electromagnetic flowmeter detects voltage generated by conductive fluid moving across a magnetic field; direction depends on the geometry of field, flow and electrodes.
- Doppler estimates velocity from frequency shift and insonation angle. Converting velocity to volume flow also requires a cross-sectional area estimate. An angle near 90 degrees makes axial Doppler sensitivity poor.

CHECK YOURSELF

1 Why may anemia increase a flow murmur despite reduced blood viscosity?

Lower viscosity raises Reynolds number, and compensatory increased cardiac output can raise velocity. Both can favor disturbed/turbulent flow. A murmur does not automatically prove a structurally abnormal valve.

SOURCES Cairculatory System 2.pptx - Slides 10-16 · Cairculatory System 3.pptx - Slides 4, 7, 13-17 · Guyton and Hall Textbook of Medical Physiology (2026).pdf - PDF pp. 178-184

SECTION 07 · LECTURE + BOOK

Measurement principles and regional circulations

A pressure transducer converts deformation into an electrical signal. Calibration, a correctly chosen reference height and a fluid system without major damping/resonance are essential: a precise number is not necessarily an accurate pressure. Electromagnetic flow measurement uses the voltage induced as conductive blood moves through a magnetic field. Doppler estimates velocity from frequency shift; beam angle matters. Volume flow requires velocity and vessel cross-sectional area, so a velocity measurement is not automatically a cardiac-output measurement.

The Fick principle obtains flow from conservation of a transported substance. For oxygen, CO equals oxygen consumption divided by arterial-minus-mixed-venous oxygen content. If oxygen consumption is 250 mL/min and the difference is 5 mL/dL, output is 50 dL/min, or 5 L/min. A pulmonary-artery sample represents mixed systemic venous return better than a random peripheral vein. Indicator-dilution and thermodilution methods infer flow from how a known indicator is diluted over time; recirculation and technique affect interpretation.

BED	DOMINANT DEMANDS AND CONTROLS	CONTRAST THAT CAN BE EXAMINED
Pulmonary	Receives essentially full RV output at low resistance/pressure; recruitment and distension accommodate flow	Local alveolar hypoxia constricts nearby pulmonary vessels, unlike hypoxic dilation in many systemic tissues
Cerebral	Neuronal metabolic coupling and strong sensitivity to CO ₂ /H ⁺ ; autoregulation within limits	Hypercapnia tends to increase flow; falling perfusion below autoregulatory capacity still produces ischemia
Renal	Large flow supports filtration as well as tissue metabolism; myogenic and tubuloglomerular feedback stabilize function	High flow does not mean the kidney necessarily has the highest oxygen extraction fraction
Skin	Sympathetic control, thermal state and arteriovenous anastomoses permit very variable flow	Heat transfer can dominate local flow requirements even when skin metabolic demand is modest
Skeletal muscle	Low resting flow can increase greatly with metabolic demand; contractions intermittently compress vessels	Local dilation can overcome much sympathetic constriction in active tissue
Coronary	Already high resting oxygen extraction; higher demand mainly requires higher flow	LV systolic compression makes diastolic supply especially important

CHECK YOURSELF

1 Doppler velocity doubles while vessel diameter halves. Does volume flow double?

No. Area becomes one quarter, and flow equals velocity times area. With uniform mean-velocity assumptions, the resulting flow is one half of the original.

SOURCES Cairculatory System 3.pptx - Flow, resistance, energy and measurement · Cairculatory System 5.pptx - Regional circulation and organ-flow examples · Guyton and Hall Textbook of Medical Physiology (2026).pdf - Chapters 14, 17, 20-21; PDF pp. 177-186, 210-221, 250-275

SECTION 08 · LECTURE + BOOK

Compliance, pulse pressure, MAP and measurement

Compliance is $\Delta V/\Delta P$. Distensibility is fractional volume change per pressure change, $\Delta V/(V\Delta P)$, so compliance equals distensibility times initial volume. Veins are highly compliant volume reservoirs. Arterial elastic recoil sustains diastolic flow after ventricular ejection. Pulse pressure is systolic minus diastolic pressure and approximately rises with stroke volume and falls with arterial compliance, although ejection speed and wave reflection also matter.

For a resting rate, MAP is often estimated as DBP plus one-third pulse pressure. For 120/80 mmHg this is about 93 mmHg, not 100 exactly. The arithmetic average is wrong because diastole lasts longer. At high heart rates or unusual waveforms, the approximation is less accurate and waveform integration is preferable. Both systolic and diastolic pressures reflect multiple cardiac and vascular factors; they are not isolated tests of heart health versus vessel health.

A cuff above systolic pressure occludes arterial flow. With slow deflation the first Korotkoff sound usually estimates systolic pressure, and disappearance estimates adult diastolic pressure in routine conditions. Palpation alone estimates systolic pressure because it detects reappearance of a pulse, not the transition to silent flow. Cuff size, arm height, rest, movement and deflation speed matter. An arm below heart level produces a hydrostatic overestimate.

CHECK YOURSELF

- 1 **Two patients have the same stroke volume but one has much stiffer large arteries. Predict pulse pressure and explain.**

Reduced compliance means the same added volume produces a greater pressure rise, so pulse pressure tends to be wider. This does not require a larger stroke volume. Wave reflection can further augment the difference.

SOURCES Cairculatory System 3.pptx - Slides 8-17 · Guyton and Hall Textbook of Medical Physiology (2026).pdf - PDF pp. 187-196

SECTION 09 · LECTURE + BOOK

Venous return, capacitance, posture and cardiac output

At steady state cardiac output equals venous return. Venous return can be modeled as (mean systemic filling pressure - right atrial pressure)/resistance to venous return. Increasing right atrial pressure tends to oppose return if other variables are unchanged, while increasing systemic filling pressure favors return. Venoconstriction recruits blood from an unstressed reservoir into stressed volume and can raise systemic filling pressure. Infusion can also shift the venous return curve, whereas hemorrhage shifts it in the opposite direction.

The actual operating point is the intersection of cardiac-function and venous-return curves. Increased contractility can increase output while lowering right atrial pressure. Pump failure often reduces output and raises right atrial pressure. A very negative right atrial pressure eventually collapses susceptible great veins, limiting further return. More suction does not produce unlimited flow.

Standing adds hydrostatic pressure below the heart and subtracts it above. Venous pooling lowers central blood volume until muscle pump, respiratory pump, venous valves and reflex venoconstriction compensate. The pressure increment depends on vertical height, not whether a person is at sea level. Atrial stretch can trigger Bainbridge tachycardia and promotes natriuretic peptide release and reduced water retention. Frank-Starling is an intrinsic length-force mechanism, not itself the Bainbridge neural reflex.

CHECK YOURSELF**1 Why can a failing heart have elevated venous pressure but low forward output?**

High atrial pressure reflects impaired emptying and backward pressure. It also reduces the systemic-to-atrial driving gradient for venous return. Filling pressure and effective output therefore need not move together.

SOURCES Cairculatory System 4.pptx - Slides 12-15 · Cairculatory System 5.pptx - Slides 19-21 · Guyton and Hall Textbook of Medical Physiology (2026).pdf - PDF pp. 250-263

SECTION 10 · BOOK EXTENSION

Venous-return curves, transmural pressure and feedback timing

At a steady operating point cardiac output equals venous return. A cardiac-function curve and a venous-return curve must be solved together. Mean systemic filling pressure is the zero-flow intercept of the return relation; resistance to venous return determines its slope. Increasing resistance flattens the curve without necessarily moving the intercept. Increasing stressed volume or venous tone usually raises the intercept. At very negative right-atrial pressure, collapse of compliant veins creates a plateau, so extrapolating the straight line predicts an impossible unlimited return.

Transmural filling pressure is intracavitary pressure minus surrounding pressure. Raising pressure around the heart by 6 mmHg requires a 6-mmHg rise inside to preserve the same distending pressure. Thus positive-pressure ventilation can raise measured central venous pressure while reducing effective filling and venous return. These effects depend on lung mechanics, blood volume and ventricular function; CVP alone is not a direct measure of preload volume.

Loss of widespread sympathetic tone expands venous capacity and lowers filling pressure generated by the same blood volume. Loss of cardiac sympathetic stimulation may simultaneously lower the cardiac-function curve. A large arteriovenous fistula initially lowers circuit resistance and rotates the return curve upward; later reflex/renal-volume adaptations may raise filling pressure. Immediate resistance effects and later volume effects are distinct. After a transfusion, redistribution and vascular stress-relaxation can partly reverse the initial increase in output.

Baroreceptors buffer rapid variation but partly reset during sustained mechanical pressure change. Electrical afferent stimulation bypasses mechanical transduction and need not reproduce receptor resetting. Low-pressure atrial/vascular stretch signals promote renal volume excretion through neural and hormonal paths. Renal pressure-natriuresis equilibrium occurs where excretion matches intake: changing intake or shifting the excretion curve changes the eventual equilibrium pressure. A volume-driven rise in pressure can later show near-normal output with higher peripheral resistance because tissues autoregulate.

CHECK YOURSELF

- 1 **A supplied model has $CO = 2 + PRA$ and $VR = 5 - 0.5PRA$, with flows in L/min and pressure in mmHg. Where is equilibrium?**

Equate the flows: $2 + PRA = 5 - 0.5PRA$. Therefore $PRA = 2$ mmHg and both flows equal 4 L/min. Neither the heart curve nor the vascular curve alone determines the answer.

- 2 **Why can a spinal sympathetic block reduce output without blood leaving the circulation?**

Loss of venous tone increases unstressed capacity and reduces mean systemic filling pressure; effective circulating filling is reduced even when total blood volume is unchanged. Concurrent reduction of cardiac sympathetic support can deepen the fall.

SOURCES Guyton and Hall Textbook of Medical Physiology (2026).pdf - PDF pp. 233-263

SECTION 11 · LECTURE + BOOK

Microcirculation, Starling forces, lymph and edema

Diffusion dominates exchange of gases and many small solutes. Lipid-soluble substances cross endothelial cell membranes readily, whereas hydrophilic solutes use clefts/fenestrae or transport pathways depending on the organ. Bulk fluid movement is governed by hydrostatic and oncotic forces plus filtration coefficient and barrier properties. It should not be confused with molecular diffusion.

The conventional Starling expression is $J_v = K_f[(P_c - P_i) - \sigma(\pi_c - \pi_i)]$. Capillary hydrostatic pressure promotes outward filtration, interstitial pressure opposes it when positive, plasma oncotic pressure opposes filtration and interstitial oncotic pressure favors it. The reflection coefficient describes restriction of protein movement. In glycocalyx-aware physiology, the effective oncotic gradient is across the endothelial surface layer and subglycocalyx space. Do not assume every systemic capillary routinely reabsorbs large amounts of fluid along its venous end; lymphatic return is essential.

KEY IDEA

Conventional course model versus research refinement

The conventional equation is the course/book calculation model. The glycocalyx and mainly outward steady-state exchange qualification is additionally supported by Levick and Michel, Microvascular fluid exchange and the revised Starling principle (2010), DOI 10.1093/cvr/cvq062, <https://pubmed.ncbi.nlm.nih.gov/20200043/>. Transient absorption and specialized absorbing beds remain possible; this is not a claim that reabsorption never occurs.

CHANGE	EFFECT ON FLUID BALANCE	EXAMPLE MECHANISM
Higher venous/capillary pressure	More filtration	Venous obstruction or heart failure
Lower plasma albumin	Less oncotic opposition	Protein loss or reduced synthesis
Higher permeability/K_f or lower protein reflection	More fluid/protein escape	Inflammatory barrier disruption
Lymphatic obstruction	Reduced protein/fluid clearance	Nodal disruption or lymphatic disease

CHANGE	EFFECT ON FLUID BALANCE	EXAMPLE MECHANISM
Sodium/water retention	Greater extracellular volume and capillary pressure	Renal/endocrine contribution

Initial lymphatic endothelial overlaps function as entry flaps anchored to surrounding tissue. Collecting lymphatics have valves and smooth muscle, assisted by skeletal and respiratory movement. Lymph returns filtered protein as well as water. Increased lymph flow, interstitial pressure changes and protein washout can buffer edema until their capacity is exceeded.

CHECK YOURSELF

1 Why can edema occur without an elevated arterial pressure?

Venous pressure, low plasma protein, increased permeability or impaired lymph drainage can each shift fluid balance. Arterial pressure is only one upstream determinant of capillary filtration.

SOURCES Cairculatory System 4.pptx - Slides 4-11 · Guyton and Hall Textbook of Medical Physiology (2026).pdf - PDF pp. 197-209 · Junqueira's Basic Histology 16th Edition.pdf - PDF pp. 237-244

SECTION 12 · LECTURE + BOOK

Local blood-flow control, autoregulation and special circuits

Active hyperemia is increased flow during increased tissue metabolism. Reactive hyperemia is transient excess flow after release of occlusion because vasodilator metabolites and oxygen debt accumulated during ischemia. Myogenic autoregulation responds to increased vessel stretch with contraction, stabilizing flow despite a changed perfusion pressure. Metabolic autoregulation links local flow to nutrient demand and waste accumulation. These mechanisms interact and have finite operating ranges.

Reduced local oxygen and increased CO₂/H⁺, K⁺, adenosine and other metabolites generally dilate systemic tissue resistance vessels. Endothelial shear stress stimulates nitric oxide, which activates smooth-muscle cGMP and relaxation. Endothelin is a potent constrictor. Histamine and bradykinin can promote dilation and increased permeability. Epinephrine is receptor- and dose-dependent: beta-2 effects can dilate some beds while alpha effects constrict others. It must not be classified as universally vasoconstricting.

The brain receives roughly 14% of a typical resting output in the lecture/Guyton illustration, not the 40% ASR wording. Cerebral flow responds strongly to CO₂/H⁺. Coronary flow follows myocardial metabolism. Renal flow has filtration-related requirements rather than simply matching oxygen use. Hepatic flow is dual, from portal vein and hepatic artery. Pulmonary vessels constrict with alveolar hypoxia, the important opposite response that redistributes flow away from poorly ventilated lung units.

CHECK YOURSELF

1 A cuff briefly stops forearm flow; release gives a transient overshoot. Why is this not the same as active hyperemia?

It is reactive hyperemia caused by the preceding interruption of perfusion and accumulated local metabolic signals. Active hyperemia is driven by increased ongoing tissue activity even without prior occlusion.

SOURCES Cairculatory System 4.pptx - Slides 16-17 · Cairculatory System 5.pptx - Slides 4-13 · Guyton and Hall Textbook of Medical Physiology (2026).pdf - PDF pp. 210-221, 264-275

SECTION 13 · LECTURE + BOOK

Baroreceptors, chemoreceptors and central pressure regulation

Carotid sinus and aortic arch baroreceptors sense stretch. Increased pressure increases afferent firing through Hering/glossopharyngeal and vagal pathways to the nucleus tractus solitarius. The integrated response decreases sympathetic tone and increases cardiac vagal activity, reducing rate, contractility and vascular resistance. A fall in pressure produces the opposite response. Rapid changes strongly influence firing. Baroreceptors buffer acute changes and can reset with sustained pressure changes.

Carotid and aortic bodies are peripheral chemoreceptors sensing oxygen, CO₂ and H⁺. Their contribution to ventilation and circulation depends on the context, especially whether breathing is allowed to change. In a low-perfusion state they can contribute sympathetic vasoconstriction. The slide statement that a generic chemoreflex simply lowers pressure should not replace this integrated account. Carotid sinus is a pressure sensor and carotid body a chemical sensor despite their proximity.

CNS ischemic responses can strongly activate sympathetic output when medullary perfusion becomes critically low. Raised intracranial pressure can elicit a Cushing response with increased systemic pressure and reflex bradycardia. Vasovagal syncope instead combines inappropriate vasodilation with reduced heart rate, lowering cerebral perfusion. Sympathetic tone is not switched off everywhere during normal rest, and an anatomic diagram of several medullary groups does not mean each group is an isolated on/off switch.

CHECK YOURSELF

1 Why does standing normally cause a rapid increase in heart rate before renal compensation could act?

Venous pooling reduces central filling and arterial stretch at baroreceptors. Reduced afferent firing rapidly increases sympathetic and reduces vagal cardiac activity. Renal volume regulation acts over a longer timescale.

SOURCES Cairculatory System 4.pptx - Slides 18-20 · Cairculatory System 5.pptx - Slides 14-21 · Guyton and Hall Textbook of Medical Physiology (2026).pdf - PDF pp. 222-232

SECTION 14 · BOOK EXTENSION

Renal-body fluid feedback and hormonal pressure control

Long-term pressure control depends strongly on sodium and water balance. Pressure natriuresis and diuresis increase salt/water excretion when renal perfusion pressure rises, reducing extracellular volume, venous return and ultimately output. The relation can shift with renal disease, sympathetic input and hormones. A single elevated reading does not establish a chronic set point or identify its mechanism.

Reduced renal perfusion, lower macula-densa salt delivery and beta-1 sympathetic drive stimulate renin. Renin cleaves angiotensinogen, and ACE generates angiotensin II. Angiotensin II constricts resistance vessels and stimulates aldosterone, thirst and vasopressin-related water conservation. Aldosterone increases distal sodium retention. Natriuretic peptides oppose volume expansion by promoting natriuresis and modulating renal/vascular function. Vasopressin increases collecting-duct water permeability and, at sufficiently high concentrations, can constrict vessels.

Do not treat hypertension as a permanent rise in resistance alone. Acute vasoconstriction can raise pressure, but chronic behavior depends on interaction with renal excretion and volume. Conversely, compensatory fluid retention after pump failure can restore some output while worsening congestion.

Mechanistic questions often ask whether an intervention changes resistance, volume, contractility or more than one variable.

CHECK YOURSELF**1 Why does an acute fall in pressure activate both a rapid neural response and slower renin-angiotensin/volume responses?**

The neural response rapidly redistributes blood and supports pressure, while endocrine and renal mechanisms conserve sodium/water to restore filling. Their timescales and controlled variables differ, so one does not replace the other.

SOURCES Guyton and Hall Textbook of Medical Physiology (2026).pdf - PDF pp. 233-249 · Cairculatory System 5.pptx - Slides 11, 19-21

SECTION 15 · LECTURE + BOOK

Coronary perfusion, exercise and oxygen demand

Left ventricular coronary flow is strongly reduced during systole by compression of intramyocardial vessels, especially in subendocardial layers, and is greatest during diastole. Right ventricular flow is less suppressed because RV systolic pressure is lower. Coronary perfusion pressure is often approximated by aortic diastolic pressure minus LV end-diastolic pressure. Tachycardia both increases oxygen demand and shortens diastolic perfusion time.

Myocardium extracts a high fraction of delivered oxygen at rest. Increased demand therefore requires increased coronary flow more than a large extraction reserve. Increased heart rate, contractility and wall stress raise oxygen consumption. With a thin-wall model, tension rises with pressure and radius, while increased wall thickness lowers stress. Dilated ventricles can require more oxygen for the same external work. This links Laplace reasoning to ischemia and failure.

Dynamic exercise combines increased sympathetic cardiac drive, increased venous return and metabolic vasodilation in active muscle. Splanchnic/renal vasoconstriction and venoconstriction help redistribute flow, while skeletal muscle pump and respiration assist return. Local metabolic dilation in working muscle can override sympathetic constriction. Total peripheral resistance may decrease even while systolic pressure and cardiac output rise. A sustained contraction can mechanically compress muscle vessels, with increased flow during relaxation.

CHECK YOURSELF**1 Why is subendocardium particularly vulnerable when tachycardia coexists with high LV filling pressure?**

Tachycardia shortens diastole, while high LV end-diastolic pressure reduces the coronary driving gradient. Subendocardial vessels already undergo strong systolic compression, so their perfusion reserve is limited.

SOURCES Cairculatory System 6.pptx - Slides 4-14 · Guyton and Hall Textbook of Medical Physiology (2026).pdf - PDF pp. 264-275

SECTION 16 · LECTURE + BOOK

Shock, heart failure and valve lesions: integrated physiology

Shock is inadequate effective tissue perfusion, not simply a low cuff pressure. Hypovolemic shock reduces circulating volume and preload. Cardiogenic shock reflects inadequate pump function. Obstructive shock impedes filling or outflow, as in tamponade or major pulmonary vascular obstruction. Distributive shock lowers effective vascular tone and can involve maldistributed flow. The early hemodynamic pattern depends on cause and stage.

Hemorrhage reduces systemic filling pressure and venous return. Baroreflex activation causes tachycardia, increased sympathetic contractility and vasoconstriction, with cold skin and reduced renal flow in a typical compensated pattern. Renin-angiotensin, vasopressin and fluid shifts contribute later. Do not memorize the lecture assertion that a fixed percentage of bleeding inevitably causes irreversible death or that contractility never increases. Outcome depends on rate, severity, compensation and treatment.

LESION	PRESSURE/VOLUME BURDEN	TYPICAL AUSCULTATORY TIMING
Aortic stenosis	LV pressure overload and increased afterload	Systolic ejection murmur
Aortic regurgitation	Diastolic aorta-to-LV backflow, volume load	Early diastolic murmur
Mitral stenosis	Elevated LA/pulmonary venous pressure, impaired LV filling	Diastolic murmur
Mitral regurgitation	Systolic LV-to-LA backflow, LA/LV volume load	Usually holosystolic murmur

Left-sided failure raises pulmonary venous pressure and can produce pulmonary edema. Right-sided failure raises systemic venous pressure and can produce peripheral edema/hepatic congestion. Reduced EF is one failure phenotype; stiff ventricles can fail despite preserved EF. S3 can reflect rapid filling of a volume-loaded ventricle, while S4 reflects atrial contraction against reduced ventricular compliance. These are physiologic associations, not sufficient diagnostic criteria alone.

CHECK YOURSELF

1 Why can tamponade resemble right-sided failure with neck-vein distension but involve neither an intrinsic RV muscle defect nor a blocked tricuspid valve?

External pericardial pressure impedes diastolic filling. Upstream venous pressure rises while effective forward output falls. The limiting structure is the pericardial enclosure.

SOURCES Cairculatory System 6.pptx - Slide 15 · Guyton and Hall Textbook of Medical Physiology (2026).pdf - PDF pp. 276-307

SECTION 17 · BOOK EXTENSION

Preserved ejection fraction, valve geometry and shock mechanisms

Ejection fraction is a fraction, not a flow. With EF 60%, EDV 120 mL gives SV 72 mL, whereas EDV 70 mL gives SV 42 mL. At 80/min their outputs are 5.76 and 3.36 L/min. A stiff small ventricle may therefore empty a normal fraction yet provide inadequate stroke volume at high filling pressure. Resting output can be maintained through elevated filling pressure while exercise reserve remains poor. Further salt/water retention eventually raises congestion more than useful output.

Aortic stenosis imposes pressure overload; LV pressure must exceed aortic pressure sufficiently to drive flow across the narrowed orifice. Regurgitation imposes a volume load and makes total ejection larger than effective forward output. If total ejection is 160 mL and regurgitation is 100 mL, forward SV is only 60 mL; at 75/min forward output is 4.5 L/min. Functional mitral regurgitation can arise from annular dilation, papillary displacement and leaflet tethering without a primary leaflet tear. S4 requires organized atrial contraction into a stiff ventricle, so atrial fibrillation can abolish the sound without curing stiffness.

Shock concerns inadequate tissue perfusion relative to need. Compensatory vasoconstriction may preserve arterial pressure better than output, masking regional underperfusion. Hemorrhage lowers stressed volume; venodilation can mimic its filling-pressure effect without identical blood loss. Plasma loss can increase hematocrit/viscosity while reducing circulating volume. Histamine-mediated venodilation plus capillary leak combines increased capacity with actual intravascular depletion.

In septic/distributive shock, low systemic resistance and normal or high total output do not prove adequate microvascular oxygen delivery. Heterogeneous flow, leakage, increased demand and microthrombi can coexist. Widespread intravascular coagulation may obstruct microvessels while consuming platelets/coagulation factors and causing bleeding elsewhere. Ischemic endothelial injury increases permeability, which worsens circulating volume and perfusion: a positive-feedback loop. Late mitochondrial injury and loss of high-energy reserves mean transiently restored pressure does not instantly reverse cellular damage. This is a mechanism of severe illness, not a bedside rule for declaring treatment futile.

CHECK YOURSELF**1 Why is high cardiac output compatible with septic shock while low output is typical in major hemorrhage?**

Sepsis can lower resistance and raise metabolic demand while distributing flow inadequately at the microvascular level. Hemorrhage usually limits filling and output directly. The label shock refers to the tissue-perfusion problem, not one obligatory cardiac-output value.

SOURCES Guyton and Hall Textbook of Medical Physiology (2026).pdf - PDF pp. 276-307

SECTION 18 · LECTURE + BOOK

ECG calibration, lead geometry and a systematic interpretation

An ECG records differences in extracellular electrical potential. A depolarization wave moving toward a positive electrode produces an upward deflection; repolarization has opposite electrical polarity. A lead is a viewing axis and is not identical to an electrode. Twelve conventional leads use ten electrodes. Limb lead I is LA-RA, II is LL-RA and III is LL-LA, so $II = I + III$. Augmented leads and precordial leads provide additional views.

CALIBRATION	SMALL SQUARE	LARGE SQUARE
25 mm/s paper speed	0.04 s	0.20 s
50 mm/s paper speed	0.02 s	0.10 s
10 mm/mV voltage gain	0.1 mV vertically	0.5 mV vertically

Before diagnosing, confirm patient/context, artifact, speed, gain and lead placement. Then assess rate, regularity, P-wave behavior, P-QRS relationship, PR, QRS width, axis, R-wave progression and ST-T changes. At 25 mm/s a regular rate is 300 divided by large squares between R waves or 1500 divided by small squares. For an irregular rhythm, count QRS complexes in a known time interval and scale to one minute. A short snapshot can misrepresent variability.

V1 and V2 lie in the fourth intercostal spaces at right/left sternal borders. V4 is fifth intercostal space at left midclavicular line. V3 lies between V2 and V4. V5 and V6 are horizontally level with V4 at anterior and midaxillary lines. The transcript phrase putting V6 in the sixth intercostal space is erroneous. Chest leads should not be moved upward to follow breast contour.

CHECK YOURSELF

- 1 **A trace is too compressed horizontally to distinguish atrial activity during tachycardia. Which adjustment improves temporal separation, and what must then change in rate calculation?**

Increasing paper speed, for example from 25 to 50 mm/s, spreads events horizontally. Each small square then represents 0.02 rather than 0.04 s. Gain changes vertical amplitude, not horizontal time.

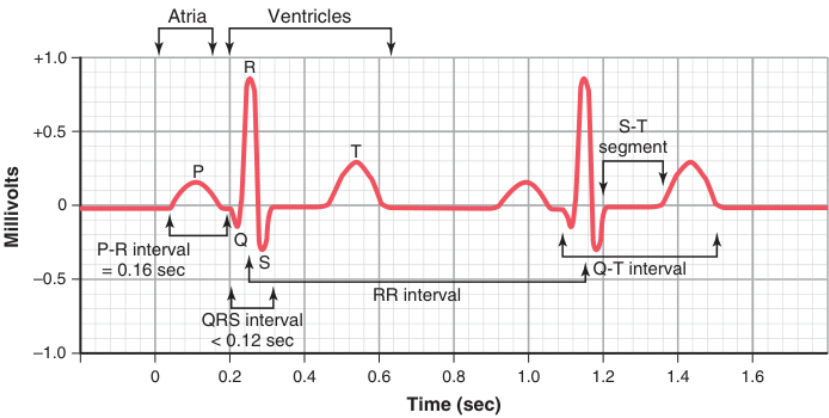


Figure 11.1 Normal electrocardiogram.

Figure 4 Distinguish waves, segments and intervals; atrial and ventricular timing is not the same as mechanical pressure.

Identify P wave; PR interval; QRS; ST segment; QT interval

Source: Guyton 2026, Fig. 11.1, PDF p. 143

SOURCES ECG Eng narration.ppt - Slides 1-64 · ECG & vectorial analysis with narration - Copy - Readable copy.pptx - Slides 1-13 · Guyton and Hall Textbook of Medical Physiology (2026).pdf - PDF pp. 142-164

SECTION 19 · LECTURE + BOOK

Cardiac axis and vector interpretation

The frontal QRS axis reflects the net ventricular depolarization vector. Lead I points to 0 degrees, II to +60, aVF to +90, III to +120, aVL to -30 and aVR to -150. An isoelectric/equiphasic QRS suggests the mean vector is approximately perpendicular to that lead. There are always two opposite perpendicular possibilities; polarity in another lead selects the correct one.

For an equiphasic aVR, candidate directions are about -60 and +120 degrees. Positive lead I and negative inferior leads favor -60, while negative I and positive inferior leads favor +120. Do not infer one number from aVR alone. The common adult normal range is approximately -30 to +90 degrees, with variation by age, body habitus and reference. Positive I and negative aVF can still be normal if lead II is positive; this distinguishes a mildly leftward normal axis from more definite left-axis deviation.

PATTERN	INTERPRETIVE STEP	PITFALL
I positive, aVF positive	Usually normal quadrant	Still inspect morphology and clinical setting
I positive, aVF negative	Check II to distinguish -30 boundary	Not every such tracing is pathological LAD
I negative, aVF positive	Rightward quadrant	Consider age, RV load, conduction or lead reversal
I negative, aVF negative	Extreme axis quadrant	Check technical errors and ventricular rhythms

An abnormal axis can reflect ventricular mass, conduction sequence, infarcted tissue or lead misplacement. QRS width helps distinguish a conduction abnormality from an isolated shift in mean direction. Ventricular repolarization often gives a positive T wave in the same lead as a positive QRS because repolarization usually propagates in the opposite spatial direction and has opposite electrical polarity.

CHECK YOURSELF

1 **Why is a positive T wave not proof that repolarization travels in the same direction as depolarization?**

Repolarization reverses the sign of the electrical wave. A spatial direction opposite to depolarization can therefore produce the same positive surface deflection.

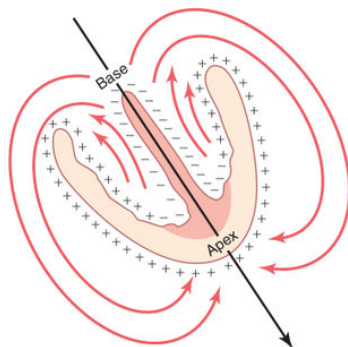


Figure 12.1 Mean vector through the partially depolarized ventricles goes from the base of the left ventricle toward the apex.

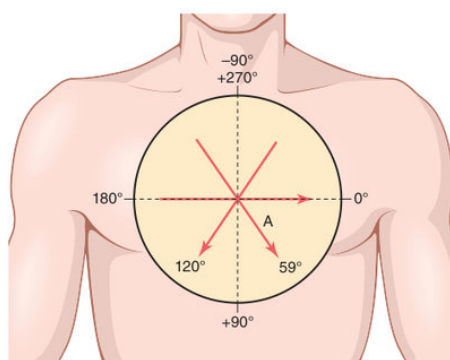


Figure 12.2 Vectors drawn to represent potentials for several different hearts and the axis of the potential (expressed in degrees) for each heart.

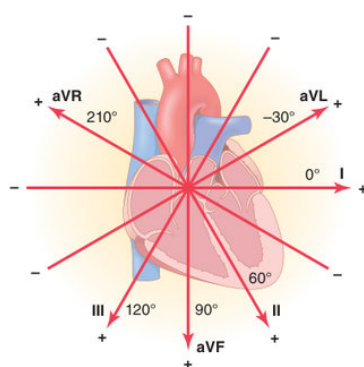


Figure 12.3 Axes of the three bipolar and three unipolar leads.

Figure 5 Use the lead axes to project a vector and resolve the two perpendicular alternatives to an equiphasic lead.

Identify Lead I; Lead II; Lead III; aVR; aVL; aVF

Source: Guyton 2026, Figs. 12.1-12.3, PDF p. 151

SOURCES ECG & vectorial analysis with narration - Copy - Readable copy.pptx - Slides 4-13 · Guyton and Hall Textbook of Medical Physiology (2026).pdf - PDF pp. 150-164

SECTION 20 · LECTURE + BOOK

ECG rhythm, intervals and difficult distinctions

PATTERN	CORE ECG REASONING	IMPORTANT DISTINCTION
Sinus rhythm	Consistent sinus P before each QRS, stable PR and appropriate P polarity	Rate can be slow or fast while origin remains sinus
Atrial fibrillation	No consistent organized P waves and irregularly irregular ventricular response	Do not diagnose from irregularity alone if artifact or ectopy explains it
Atrial flutter	Organized repetitive atrial activity, often sawtooth	Ventricular response may be regular with fixed block
First-degree AV block	All P waves conducted with prolonged PR	Delay rather than dropped beats
Mobitz I	Progressive PR prolongation before a dropped QRS	Group beating needs enough strip length
Mobitz II	Intermittent nonconducted P with otherwise stable PR	A 2:1 pattern often cannot be reliably subtyped from a short strip alone
Complete AV block	Atrial and ventricular rhythms independent	AV dissociation requires analysis of both rhythms
Ventricular ectopic/tachycardic pattern	Wide abnormal QRS and altered activation sequence	Wide QRS can also arise supraventricularly with aberrancy

PR is normally about 0.12-0.20 s in adult teaching examples. A QRS of at least 0.12 s is wide. QT depends on rate, so QTc interpretation requires a correction method and clinical context. ST elevation, depression or T inversion can indicate ischemia but have alternative causes; compare contiguous leads, reciprocal changes, morphology and timing. A single labeled internet example is not a universal template. Anatomical territories are broadly inferior II/III/aVF, high lateral I/aVL, and precordial septal/anterior/lateral views, with overlapping supply.

CHECK YOURSELF

1 A regular ventricular rate near 150/min appears without obvious P waves. Why should atrial flutter with 2:1 conduction remain a possibility?

One flutter wave may be buried in each QRS/T cycle. A longer strip and leads that display atrial activity better can reveal organized atrial activity around twice the ventricular rate. Rate alone cannot establish the diagnosis.

SOURCES Guyton and Hall Textbook of Medical Physiology (2026).pdf - PDF pp. 165-176 · ECG Eng Appl Physio Assignment.ppt - Slides 1-4

SECTION 21 · BOOK EXTENSION

Pacemaker competition, reentry and vector calculations

The fastest effective pacemaker normally repeatedly resets subsidiary pacemakers. After sinus-to-ventricle conduction fails, a slower escape rhythm may take time to emerge. Acetylcholine increases nodal K⁺ conductance and suppresses pacemaker/Ca²⁺ currents, hyperpolarizing cells and delaying threshold. Slower Purkinje/ventricular propagation broadens QRS and reduces synchrony, whereas AV-nodal delay mainly alters the atrial-to-ventricular interval.

Reentry requires an excitable pathway, suitable unidirectional block and sufficient time for tissue ahead of a returning wave to recover. Circuit transit time is path length divided by conduction speed. Equivalently, refractory wavelength equals conduction speed times refractory period. A 12-cm circuit at 40 cm/s has a 0.30-s transit time; with a 0.25-s refractory period, recovery before return is possible. If speed falls to 25 cm/s and refractory period rises to 0.36 s, wavelength is still only 9 cm, below circuit length. Prolonging refractoriness therefore does not automatically terminate reentry if conduction changes too.

A surface lead measures projection, not the full cardiac vector magnitude. A 2-mV vector at +60 degrees projects as $2 \cos 60 = +1$ mV in lead I. Calibration is equally essential: 8 mm horizontally at 50 mm/s is 0.16 s; 8 mm vertically at 5 mm/mV is 1.6 mV. Low surface voltage may reflect altered myocardial activation, surrounding fluid or increased lung air and is not a direct one-to-one gauge of wall thickness. Atrial repolarization occurs but is usually obscured by QRS.

CHECK YOURSELF

- Lead I is positive, aVF negative and lead II negative. What is the geometric axis interval before assigning a clinical label?**

I+/aVF- places the vector between -90 and 0 degrees. Lead II is perpendicular at -30 degrees, so its negative projection narrows the interval to approximately -90 to -30 degrees. Borderline equiphasic tracings require allowance for measurement error.

SOURCES Guyton and Hall Textbook of Medical Physiology (2026).pdf - PDF pp. 135-176

SECTION 22 · LECTURE + BOOK

Erythrocytes, hematocrit and oxygen delivery

Mature erythrocytes are flexible biconcave anucleate cells approximately 7-8 micrometers across. Their shape increases surface area relative to volume and permits capillary passage. Hemoglobin carries oxygen, carbonic anhydrase accelerates CO₂ hydration, and hemoglobin buffers H⁺. Lacking mitochondria, RBCs obtain ATP through glycolysis rather than consuming the oxygen they transport. Membrane-cytoskeletal integrity and redox protection are essential for their survival.

Hematocrit is the packed RBC fraction of whole blood volume. Hemoglobin concentration, RBC count and hematocrit are related but are not interchangeable. Anemia is principally a reduction in hemoglobin/oxygen-carrying capacity relative to the appropriate reference population, not necessarily a reduction in the number of cells. Plasma volume changes can alter measured concentration without an equivalent immediate change in total red-cell mass.

INDEX	FORMULA WITH CONVENTIONAL UNITS	INTERPRETATION
MCV (fL)	$\text{Hct (\%)} \times 10 / \text{RBC count (million}/\mu\text{L)}$	Mean cell volume

INDEX	FORMULA WITH CONVENTIONAL UNITS	INTERPRETATION
MCH (pg)	$\text{Hb (g/dL)} \times 10 / \text{RBC count (million}/\mu\text{L)}$	Mean Hb per cell
MCHC (g/dL)	$\text{Hb (g/dL)} \times 100 / \text{Hct (\%)}$	Hb concentration within packed RBC volume
O2 content (approx.)	$1.34 \times \text{Hb} \times \text{saturation} + 0.003 \times \text{PaO}_2$	mL O ₂ /dL when Hb in g/dL, saturation as a fraction, PaO ₂ in mmHg

Oxygen delivery equals cardiac output times arterial oxygen content with unit conversion. A patient can have normal PaO₂ but low oxygen content because hemoglobin is low. Increasing dissolved oxygen contributes little compared with hemoglobin-bound oxygen under ordinary conditions. Interpret oxygen pressure, saturation and content as distinct quantities.

CHECK YOURSELF

1 Why can a severely anemic patient have a normal pulse-oximeter percentage?

The percentage reports the fraction of available hemoglobin binding sites occupied, not how much hemoglobin is present. Normal saturation with low Hb can still mean low total arterial oxygen content.

SOURCES blood 1.ppsx - Slides 2-17 · Guyton and Hall Textbook of Medical Physiology (2026).pdf - PDF pp. 440-449 · Junqueira's Basic Histology 16th Edition.pdf - PDF pp. 248-255, 265-274

SECTION 23 · LECTURE + BOOK

Erythropoiesis, iron and anemia mechanisms

Low tissue oxygen availability increases renal erythropoietin production through hypoxia-responsive signaling. EPO promotes erythroid progenitor survival and differentiation in marrow. The series proceeds through proerythroblast, basophilic, polychromatophilic and orthochromatic erythroblast stages, nuclear extrusion, reticulocyte and mature RBC. Cell size and nuclear size fall while hemoglobin accumulates. Reticulocytes retain residual RNA and indicate recent marrow output, but the percentage must be interpreted relative to the anemia and maturation time.

Vitamin B12 and folate support DNA synthesis, so deficiency produces ineffective, megaloblastic erythropoiesis. Iron deficiency limits heme synthesis and commonly produces microcytic hypochromic cells. Chronic blood loss can deplete iron, while acute blood loss is initially normocytic and may not immediately lower concentration before fluid redistribution. Renal failure can reduce EPO, marrow failure can impair multiple cell lines, and hemolysis increases red-cell destruction.

Iron is absorbed mainly in the proximal small intestine, circulates bound to transferrin and is stored principally in ferritin, with hemosiderin in larger stores. Hepcidin reduces ferroportin-mediated iron export from enterocytes and macrophages, restricting availability. Senescent RBCs are removed mainly by macrophages, especially in spleen/liver/marrow; iron is recycled and heme is converted toward bilirubin. Mature RBC life is approximately 120 days. Secondary erythrocytosis can be hypoxia-driven, whereas a myeloproliferative process is not the same adaptive mechanism.

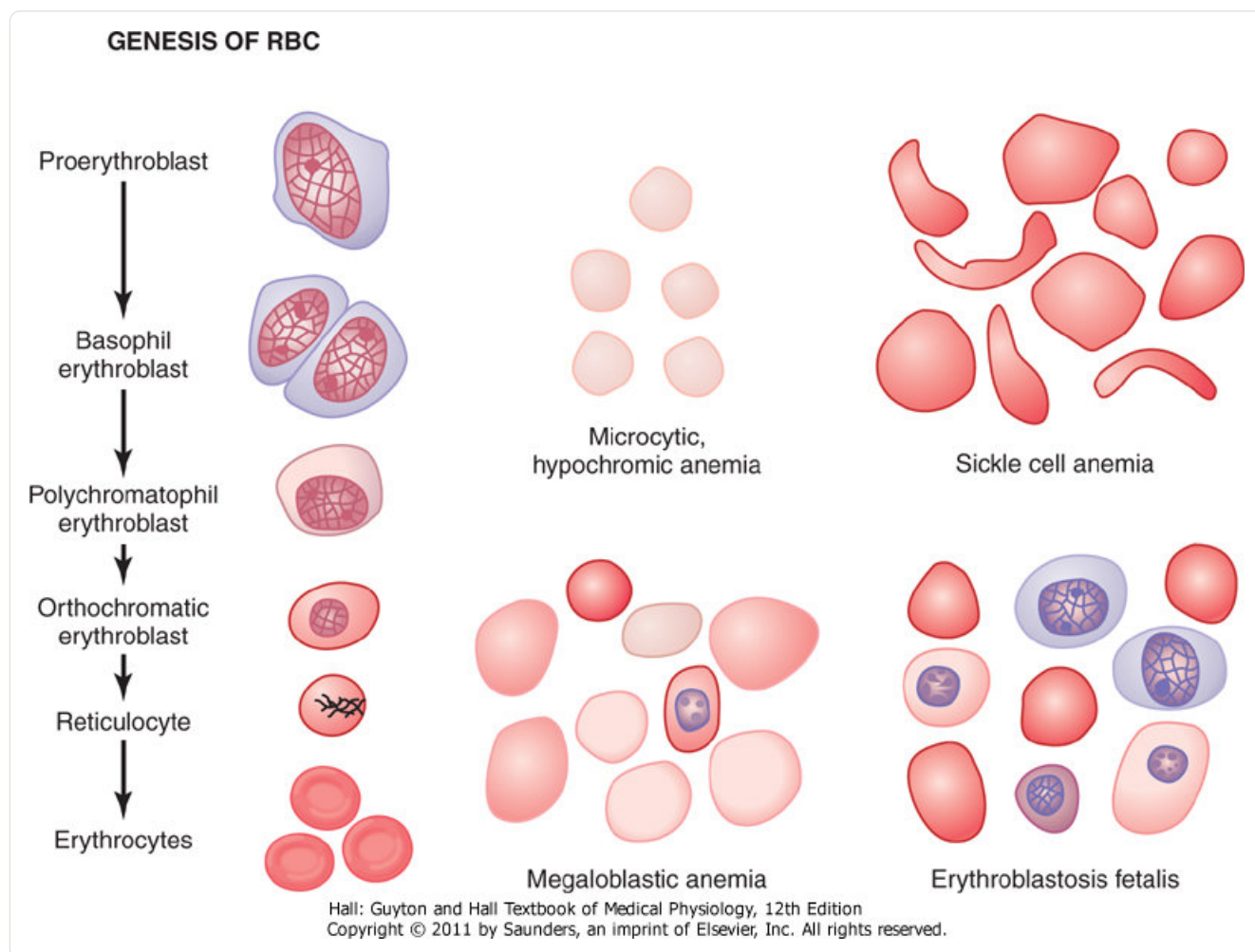


Figure 6 Follow the decrease in size, chromatin condensation, changing cytoplasmic staining and nuclear loss across erythroid maturation.

Identify Proerythroblast; Basophilic erythroblast; Polychromatophilic erythroblast; Orthochromatic erythroblast; Reticulocyte; Erythrocyte

Source: blood 1.ppsx, slide 10

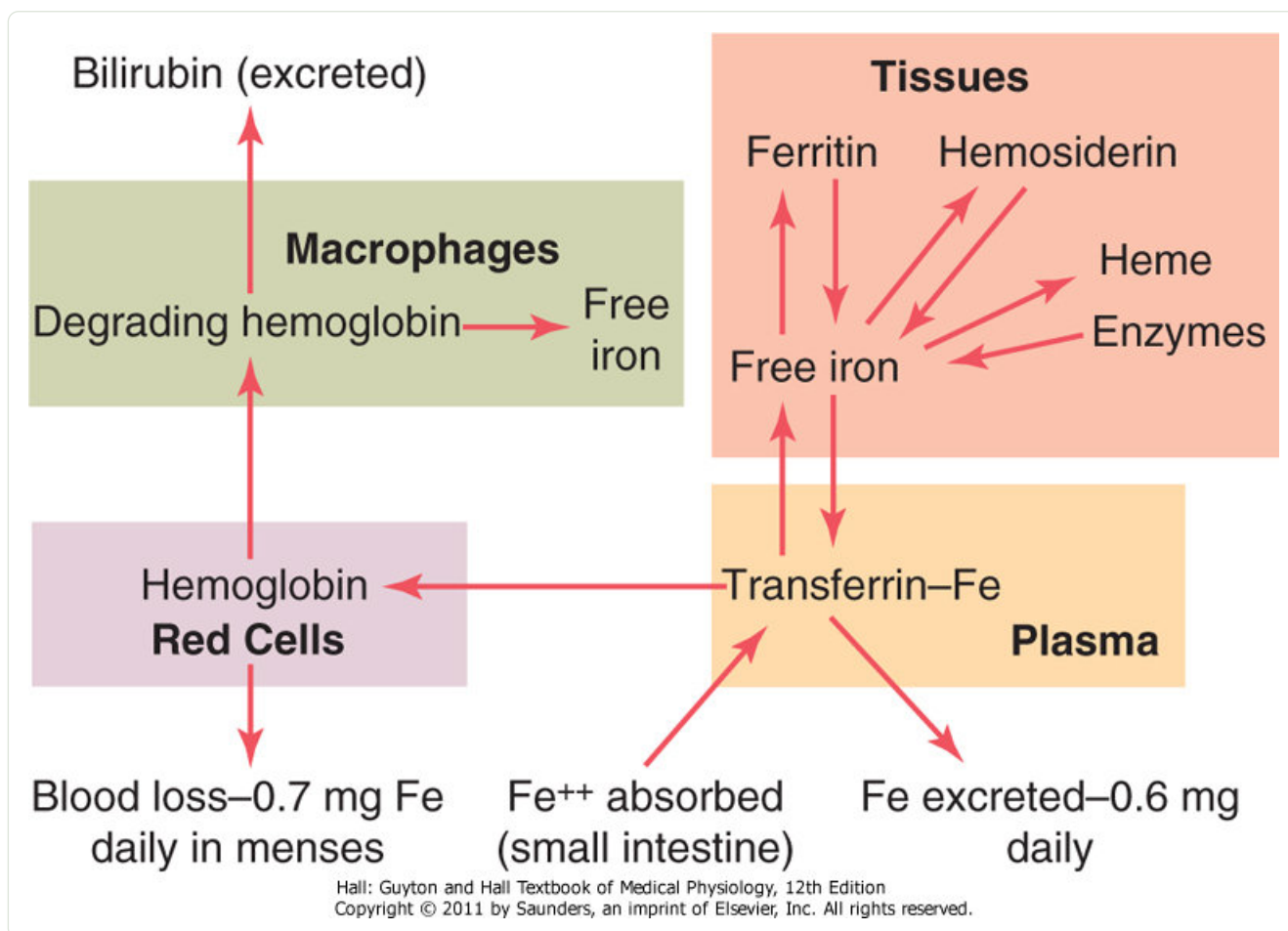


Figure 7 Connect marrow use, circulating transport, tissue storage and macrophage recycling of iron.

Identify Transferrin; Ferritin; Hemosiderin; Macrophages; Hemoglobin; Bilirubin

Source: blood 1.ppsx, slide 27

SOURCES blood 1.ppsx - Slides 8-32 · Guyton and Hall Textbook of Medical Physiology (2026).pdf - PDF pp. 440-449 · Junqueira's Basic Histology 16th Edition.pdf - PDF pp. 265-274

SECTION 24 · LECTURE + BOOK

Leukocytes, inflammation and absolute counts

CELL	MICROSCOPIC RECOGNITION	MAIN FUNCTIONAL ASSOCIATION
Neutrophil	Segmented nucleus, fine pale granules	Acute bacterial defense, phagocytosis, oxidative killing
Eosinophil	Usually bilobed nucleus, large red/orange granules	Helminth defense and allergic inflammation
Basophil	Dark coarse granules may obscure nucleus	IgE-associated mediator release
Monocyte	Large cell, folded/kidney-shaped nucleus, gray-blue cytoplasm	Circulating precursor contribution to tissue macrophage populations
Lymphocyte	Dense round nucleus and scant cytoplasm in small forms	Adaptive immune populations; NK cells are innate lymphocytes

Tissue-resident macrophages can respond immediately to injury. Neutrophils are recruited from blood, while later monocyte-derived macrophages and increased marrow production sustain responses. Selectin-mediated rolling, chemokine activation, integrin-mediated firm adhesion and transendothelial migration are sequentially different events. The principal site of routine inflammatory leukocyte exit is the postcapillary venule. Opsonins such as IgG and C3b improve phagocytosis. NADPH oxidase generates superoxide; downstream peroxide and myeloperoxidase-dependent hypochlorous species contribute to killing.

Differential percentages require the total WBC concentration. If lymphocytes are 50% of 3,000 WBC/ μL , their absolute count is 1,500/ μL . A high relative fraction alone does not establish absolute lymphocytosis. Similarly, an apparently moderate neutrophil percentage can coexist with severe absolute neutropenia when total WBC count is very low. Reference intervals vary with age and laboratory.

CHECK YOURSELF

1 Why can leukemia coexist with infection, anemia and bleeding despite a very high WBC count?

A high count may consist of dysfunctional/clonal cells and can displace normal marrow production. Effective host defense, erythropoiesis and platelet production can therefore be impaired rather than improved.

SOURCES blood 3.pptx - Slides 4-35 · Guyton and Hall Textbook of Medical Physiology (2026).pdf - PDF pp. 450-459 · Junqueira's Basic Histology 16th Edition.pdf - PDF pp. 250-263

SECTION 25 · LECTURE + BOOK

Primary hemostasis and platelet biology

Hemostasis begins with local vasoconstriction and a platelet response, then a fibrin network stabilizes the plug. Platelets are anucleate cytoplasmic fragments of megakaryocytes. Their cytoskeleton changes shape during activation, granules release mediators, and membrane phospholipids provide a reaction surface for coagulation complexes. Primary and secondary hemostasis interact; they are not isolated systems reserved exclusively for small versus large vessels.

Exposed collagen and von Willebrand factor support platelet adhesion, particularly through GPIb-vWF interaction. Activation releases ADP and produces thromboxane A₂, recruiting more platelets. Activated GPIIb/IIIa binds fibrinogen to bridge platelets in aggregation. Alpha granules contain proteins such as vWF and growth factors; dense granules contain ADP/ATP, calcium and serotonin. Cytoskeletal contraction contributes to clot retraction. The liver is a major source of thrombopoietin.

DEFECT OR PROCESS	MECHANISM	TYPICAL PATTERN TO REASON FROM
Low platelet number/function	Poor initial plug	Mucocutaneous bleeding, petechiae or prolonged bleeding from small injuries
vWF problem	Adhesion impaired and factor VIII stabilization affected	Can bridge platelet and coagulation abnormalities
GPIb defect	Adhesion receptor defect	Different from aggregation receptor failure
GPIIb/IIIa defect	Fibrinogen-mediated aggregation impaired	Adhesion and aggregation are not interchangeable

EXAM TRAP**Lifespan versus half-life**

The lecture gives an approximately 8-12 day survival figure but labels it half-life. Treat it as an approximate platelet lifespan statement, not a rigorously defined elimination half-life.

SOURCES blood 2.ppsx - Slides 2-14 · Guyton and Hall Textbook of Medical Physiology (2026).pdf - PDF pp. 480-485

SECTION 26 · LECTURE + BOOK

Coagulation cascade, anticoagulants and fibrinolysis

Tissue factor with factor VIIa initiates coagulation after injury. The classic intrinsic scheme uses XII, XI, IX and VIII; the extrinsic route is tissue factor/VII. They converge on factor X. Xa with Va, calcium and a phospholipid surface forms prothrombinase, generating thrombin from prothrombin. Thrombin converts fibrinogen to fibrin, activates factor XIII to cross-link it and amplifies activation of platelets and several cofactors. The cell-based in-vivo process is more integrated than two independent linear tubes.

COMPONENT	ESSENTIAL DISTINCTION
I / II / III / IV	Fibrinogen / prothrombin / tissue factor / calcium
V and VIII	Cofactors, not the same proteases as activated IX or X
XIII	Cross-links fibrin after fibrin formation
Vitamin K dependent	II, VII, IX, X plus proteins C and S
Antithrombin	Inhibits thrombin and activated coagulation proteases; enhanced by heparin
Protein C/S	Activated protein C with S inactivates Va and VIIa
tPA and plasmin	tPA activates plasminogen; plasmin degrades fibrin
PT versus aPTT	PT emphasizes extrinsic/common; aPTT intrinsic/common pathways

Serum is the liquid remaining after clotting and differs from plasma by consumption/removal of fibrinogen and changes in several coagulation proteins. Most clotting proteins are synthesized in liver, but the slide statement that all except tissue factor come from hepatocytes is too broad: endothelial cells are important for vWF and factor VIII. Vitamin K supports gamma-carboxylation needed for calcium-dependent membrane binding. Fibrinolysis is regulated continuously and should not be imagined as beginning only a fixed number of days after injury.

CHECK YOURSELF

1 Why can an isolated factor XIII deficiency produce bleeding while routine PT and aPTT remain normal?

Those tests detect formation of the initial fibrin clot. Factor XIII stabilizes fibrin by cross-linking after that endpoint, so defective clot stability may not prolong those screening times.

SOURCES blood 2.ppsx - Slides 15-31 · Guyton and Hall Textbook of Medical Physiology (2026).pdf - PDF pp. 483-491

SECTION 27 · BOOK EXTENSION

ABO, Rh and transfusion concepts

ABO antigens are carbohydrate determinants on RBC surfaces. Plasma commonly contains antibodies against the absent A/B antigen, with important age and immune-history exceptions. A recipient receiving incompatible donor RBCs can develop antibody-mediated agglutination and hemolysis. RBC and plasma compatibility run in different directions because one product supplies cells and the other supplies antibodies.

Rh D is a major RBC protein antigen. Anti-D generally follows exposure rather than being naturally present in an unsensitized person. Maternal IgG can cross the placenta and damage fetal RBCs if fetal antigen is incompatible with maternal sensitization. Distinguish immune hemolysis from an intrinsic membrane or hemoglobin defect. The phrase universal donor is context-dependent: component type, Rh, minor antigens and crossmatching remain important.

CHECK YOURSELF

- 1 Why does the blood group that is convenient for emergency RBC donation not automatically make the best universal plasma donor?**

RBC compatibility is governed mainly by donor cell antigens meeting recipient antibodies. Plasma compatibility depends on donor antibodies encountering recipient RBC antigens, reversing the ABO logic.

SOURCES Guyton and Hall Textbook of Medical Physiology (2026).pdf - PDF pp. 474-479